## 4.1: Popular Translation Proposal

### 3D Printing the Future

**TEAM MEMBERS:**

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**STEM-BASED TOPIC:**

3D Printing

## Proposal

For this proposal, we have chosen 3D Printing as our STEM topic. 3D printing has been around for more than 30 years, but only recently has it made its way into the public sphere. But first, what is 3D printing? 3D printing is the process of making a physical object from a three-dimensional digital design. This process is usually done by 'printing' thin layers of material over each other and fusing them. When most people think about 3D printing, they probably think about making little toys and figures to mess around with or display but its use has much more potential than that. 3D printing can be and is used in multiple different industries and applications. From healthcare, fashion, design, and even personal use, the versatility of 3D printing is immense. For example, medical and healthcare professionals sometimes use 3D printing to create custom fitted prosthetic limbs for patients. In situations like this, 3D printing is an amazing and adaptable tool that is used to improve the quality of life for many people. Despite what the specific application is, 3D printing offers a cost effective and quick production time when it comes to turning a digital concept into a physical one.

In the media, 3D printing is often depicted in movies as a fast and easy process but that is hardly the case. While, the printing process is relatively faster than most other ways of making a product, the process time is still long, ranging from an hour to a day depending on the scale and complexity of whatever is being created.

For our campaign, we would use YouTube in conjunction with other platforms to inform people about 3D printing. Creating YouTube videos would then allow us to splice those videos into shorter form content like TikTok, Instagram Reels, and YouTube Shorts. These platforms are among the most popular right now and create large engagement with people who might not even be looking into 3D printing. Short form quick informational videos would do well and help create a more informed public.

3D printers in reality have been around for several decades with the first 3D printer that was able to print an actual part created in 1983 by Chuck Hull (Adlughmin). However, 3D printers have existed in fiction for much longer. One of the earliest mentions of creating objects via machines like 3D printers was 83 years ago in a novelette titled "The 4-Sided Triangle" by William F. Temple. In this novelette, a duplicating machine is used to make copies of objects, although the method of how it works is not expanded on. In Murray Leinster's 1945 story, "Things Pass By," a man creates a spaceship one layer at a time, a first mention of creating an object by printing an object via objects ("What Is the First Reference to a 3D Printer in Science Fiction?"). However, as fiction has turned to fact, it is important to make note

of what is actually possible and what still remains outside of our, current, technological grasp. We will analyze the following texts, *Ocean's 8*, *Westworld*, and *Lost in Space*. As a side effect, spoilers for these texts may exist.

A lot of media currently is a mix of fact and fiction; the bridge between what we know is possible, but we cannot quite reach yet. One example of this is *Ocean's 8.* In *Ocean's 8*, the heist crew scans and then 3D prints a valuable necklace so as to swap it with the real necklace during a banquet scene. While 3D printing technology to replicate diamonds is in the works, the consensus is that it isn't quite possible yet. However, that didn't stop Warner Bros from trying to accurately represent the technology, working with MakerBot, a 3D printer company. "When Warner Bros approached us with the pitch for Ocean's 8, we were thrilled to introduce them to the Replicator Z18 and give our best tips for how to shoot the printer without adding too much CG. That's always a big concern — especially with the history of hype and media portrayals creating unrealistic expectations of the technology" (Goehrke). While this text had some incorrect aspects, it overall accurately represented the growth and possibilities of 3D printing as a whole. Unfortunately, viewers won't be able to complete elaborate heists anytime soon with the help of 3D printers, but *Ocean's 8* works as a good introduction to creative uses of the technology.

On the other hand, some current science-fiction shows are currently very far from feasible such as *Westworld* where bionic robot flesh is printed in mass and with rapid speed. However, while the rapid printing and scale that is seen in this show is not currently available, 3D printed muscles and organs could be much closer. Back in 2016, MIT developed a type of plastic muscle that researchers were able to control via heat (Liszewski). In addition, researchers at Columbia have also been making strides in research about artificial muscles (Evarts). While the technology doesn't exist yet, innovations in the research field are constantly improving. Overall, *Westworld* focuses more on the ethical questions posed by A.I. and the 3D printing aspect is just something that allows the A.I. to exist in our world. While neither the current A.I. nor the 3D printing technology is at the level shown in the show, it poses important questions to viewers about ethics and things that need to be considered as research continues.

Finally, *Lost in Space* shows uses of 3D printing that are not only useful, but are currently possible. Tools and medical devices such as arm braces are printed to help characters as they travel through space. This representation is not only correct, but also is something that current astronauts are using on the International Space Station where a 3D printer has been installed (Gaskill). This is a valuable depiction of 3D printing as it not only shows how 3D printing can be used on Earth today, but also shows how it could impact our future in space.

Overall, these texts have a decent audience reception. While *Ocean's 8* has a lower audience score on Rotten Tomatoes than the other two, *Lost in Space* has an average audience score of 78% and *Westworld* with a score of 72% (Rotten Tomatoes). The lower response to *Ocean's 8*, can also be attributed to being a reboot of a popular movie series with an all-female cast. In turn, the overall reception of these texts by STEM experts was positive. Some parts were met with skepticism, particularly in *Ocean's 8* where no supports were used to print the diamonds (Goehrke). However, these texts are met with suspension of disbelief and discussion of how far the technology has to go before it reaches these capabilities, for *Westworld* and *Ocean's 8*, and the accurate depiction, for *Lost in Space*. In turn, casual viewers seem to enjoy the depictions as a whole. 3D printer hobbyists commented on Reddit about the different medias and praised the accuracy of *Lost in Space.* One thread discussed how

*Lost in Space* showed depictions of 3D printed buildings in the background of one of the scenes ("r/3dprinting - Lost in Space 3D Printed Buildings.") and another thread commented on how the characters 3D printed medical braces ("r/3dprinting - 3D Printing in the New Lost in Space Show."). Although, one viewer was quick to point out that it is only polite to remove the support material when you take your finished print off of the printer! While these viewers are hobbyists instead of casual viewers, the consensus is shared that 3D printing is a cool and useful piece of technology. With our proposal we will work to reach further audiences and expand on points of disinformation and highlight scientific achievements.

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